The residual centralizer at a singleton end

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$ be a transitive string C-group, $n \geq 12$, with all maximal omissions intransitive. Suppose G_0 has orbits $\{a\}$ and B, and write the unique crossing as ab. Put $X = \Omega \setminus \{a, b\}$, $K = G_{>1}$, and $\rho_0 = (a\ b)\delta$. Then K fixes a, b, preserves X, and commutes with the odd involution δ on X. Neither transitivity nor a full-support assumption on K is made.

Lemma 1 (An actual orbit-deficit inequality)**.** *Let t be the number of two-point δ-orbits and f its number of fixed points in X. Let z be the number of orbits of $\langle K, \delta \rangle$ on X, including singleton orbits. Then*

$$r - 2 \leq t + f - z. \tag{1}$$

Moreover the distinguished images of the $r-2$ generators of K in the permutation action on the $t + f$ orbits of δ form an independent generating set.

Proof. Commutation makes the partition into δ-orbits invariant under K. For a distinguished generator $s = \rho_j$, $j \geq 2$, let K_j^- be generated by the other distinguished K generators. Suppose its image has the same orbits on the δ-blocks as the full image of K. Adjoining δ makes the point orbits of K_j^- on X exactly the unions of blocks in these block orbits: inside every two-point block δ supplies both points. Thus $\langle K_j^-, \rho_0 \rangle$ and $\langle K, \rho_0 \rangle$ have exactly the same point orbits on Ω, including the orbit $\{a, b\}$. Adjoining the same permutation ρ_1 to orbit-equivalent groups leaves them orbit-equivalent, since the resulting partition is the equivalence closure of the old partition and the edges of ρ_1. This would make G_j transitive, a contradiction. Therefore deleting any distinguished image strictly refines its orbit partition. In particular no image is generated by the others, proving independence.

Choose for each distinguished image one of its edges joining different orbits after that image is deleted. The selected edges form a forest on the $t + f$ blocks: a cycle would give an alternate path avoiding the label of any one of its edges, contradicting that edge's eligibility. The forest has at least z components because every selected edge stays inside a full block-action orbit. It has $r - 2$ edges, giving (1). □

Theorem 2 (No fixed-point-free residual exception)**.** *If δ has no fixed point on X, then $r \leq \lfloor (n-1)/2 \rfloor$.*

Proof. Here $n = 2m + 2$ with $m = t$ odd, since δ is odd. Thus $m \geq 5$. By Lemma 1 and the retained Whiston theorem, the independent pair images have length $r - 2 \leq m - 1$. Failure of the desired rank bound would force $r = m + 1$. Whiston's exact equality clause then identifies the pair image as the full S_m.

Write $\sigma = \rho_1$. The group $G_0 = \langle \sigma, K \rangle$ is transitive on B, whereas K fixes b, so $\sigma = (b\ x)v$ for some $x \in X$ and an odd involution v supported on $X \setminus \{x\}$. Set $x' = x^\delta$ and $H = G_{>2}$. Its generators commute with σ and fix b, hence fix x; they also commute with δ, hence fix x'. Their $m - 2$ pair images are independent, as a subfamily of the images in Lemma 1, and fix the pair $\{x, x'\}$. Whiston therefore makes their image the full S_{m-1} on the other pairs.

The only H-fixed points of X are x, x', because its pair image moves every other pair. The permutation v centralizes H and fixes x, so also fixes x'. On the remaining even number $m-1 \geq 4$ of pairs, the accepted paired-row centralizer lemma makes v even. Its hypotheses require the

actual full symmetric pair image and the common pair swap, not transitivity of H on points; they allow both a transitive action and two faithful rows. This contradicts the oddness of v and proves the theorem. □

The remaining critical range. Combine this theorem with the separately written odd-tail reduction, under its explicit smaller-degree alternating induction. A counterexample in the singleton-endpoint branch must have $n = 2r \geq 14$, $G_0 = A_B$, and K intransitive on its support. Its residual δ has an odd number t of transpositions and a *nonempty even* number f of fixed points. Lemma 1 gives exactly

$$n = 2 + 2t + f, \qquad r - 2 = t + f/2 - 1, \qquad z \leq f/2 + 1. \tag{2}$$

Since K preserves both the nonempty fixed-point set and the nonempty union of pairs, $z \geq 2$. In particular $f = 2$ forces $z = 2$: K is transitive on the two fixed points and on the set of t pairs, but need not be transitive on all $2t$ points in those pairs. No IP is inferred for either projected action.

Attribution and limit. The endpoint and pair-centralizer strategy is borrowed from Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.29, physical page38, and its previously accepted full replacement proof. The actual orbit-equivalence argument and inequality (1) extend that proof to an intransitive tail without a fixed-point-free hypothesis on the tail itself. Theorem 2 removes the tail-transitivity requirement while retaining the necessary fixed-point-free residual hypothesis. All retained Whiston inputs and the complete paid paired-row proof remain dependencies. This is written mathematics; no new computation, formalization or source-wide recovery is claimed. The cases $f \geq 2$, the source's further equality graphs and the complete L044 recovery remain open here.